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ON A TORSION-FREE WEAKLY BRANCH GROUP DEFINED BY A THREE STATE AUTOMATON*

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We study a torsion free weakly branch group G without free subgroups defined by a three state automaton which appears in different problems related to amenability, Galois groups and monodromy. Here and in the forthcoming paper [20] we establish several important properties of G related to fractalness, branchness, just infinitness, growth, amenability and presentations.

1. Introduction

Groups generated by finite automata were introduced in sixties and now play an important role in such topics of group theory and its applications as Burnside problem [12], groups of intermediate growth and amenable groups [13], just infinite groups [17], the Atiyah conjecture about L^2-Betti numbers [19, 22], etc. One of the interesting topics is the question which groups can be embedded in the group of finite automata.

Recently new applications of automata groups were discovered. It was realized that with their help one can compute spectra of groups and graphs [2, 19], that they appear in the study of the Galois group of the iterations of polynomials (in the case of fields of positive characteristics) or they appear as monodromy groups of normal coverings. This makes the study of such groups particularly interesting. For a

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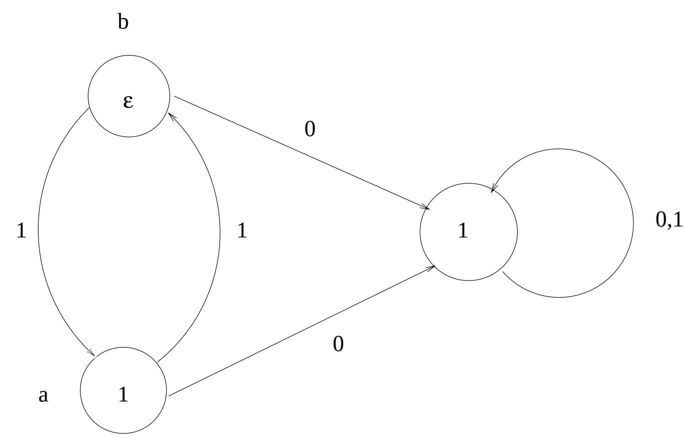

Fig. 1. The automaton A.

general information about automata and groups generated by them we recommend [10, 23].

In this paper we investigate an interesting example of a group generated by a three state automaton from Fig. 1 (all relevant definitions will be given below).

We started the study of this group in 2000 because of its interesting spectral properties [20]. Later this group appeared in connection to the Galois groups of the iterations of the polynomial x^2-1 over finite fields [32], as a monodromy group of a ramified covering of the Riemann sphere given by the polynomial z^2-1 (see [28]) and in the relation to the Julia set of the map $z \to z^2-1$ (see [1]). All these recent discoveries make the investigation of properties of the group G very important.

In this paper we prove the following

Theorem 1. *Let G be the group generated by the automaton from Fig.* 1.
The group G

(a) *is fractal*;
(b) *is regular weakly branch over G'*;
(c) *is torsion free*;
(d) *has exponential growth*;
(e) *is just non-solvable*;
(f) *has a presentation*:

$$G = \langle a, b | \sigma^{\varepsilon}(\theta^{m}([a, a^{b}])) = 1, m = 0, 1, \ldots, \varepsilon = 0, 1 \rangle \,,$$

where

$$\sigma : \begin{cases} a \to b^2 \\ b \to a \end{cases} \qquad \theta : \begin{cases} a \to a^{b^2+1} \\ b \to b \end{cases} \quad ;$$

(g) *has solvable word problem*;
(h) *is not in the class SG of subexponentially amenable groups*;

(i) *is in the class NF of groups with no free subgroup with* 2 *generators*;
(j) *is contracting.*

Definitions of fractal, contracting and weakly branch group is given in Sec. 2.

As a reference for growth we give [21] or [25]. An infinite group is just non-solvable if every proper quotient is solvable. The classes AG, EG, SG and NF are defined as follows. AG is the class of all discrete amenable groups, i.e. those groups G for which there exists a Banach mean on $\ell^\infty(G)$ [11]. EG is the class of elementary amenable groups, i.e. the smallest class containing finite and abelian groups which is closed under taking extensions, subgroups, factor groups and direct limits. NF is the class of groups which do not contain a free group of rank 2. The classes EG and NF first appeared in [8]. Finally let SG be a class of subexponentially amenable groups, i.e. the smallest class of groups containing groups of sub-exponential growth and closed under taking extensions and direct limits (for more on elementary classes see [31]).

The question whether the classes AG and EG, AG and NF coincide were asked by Day [8] and were answered by negative in [14, 29]. The same questions but for finitely presented groups were answered by negative in [16, 30]. It is still a matter of great importance to construct a new examples if finitely generated and perhaps finitely presented non-amenable groups without free subgroups of rank 2 which would satisfy a new properties (for instance finitness condition like being residually finite). It is also important to produce new examples of amenable but not elementary amenable groups. The following question was asked in [16]:

Question 1. Is it true that $SG = AG$?

Let us remark that all examples of amenable but non-elementary amenable groups constructed in [14, 16] and in other papers belong to the class SG.

Question 2. Is the group G amenable?

If the answer to the second question is positive then according to Theorem 1 the answer to Question 1 is negative and we get new examples of non-amenable groups. If the answer to Question 2 is positive we get an example of a non-amenable residually finite group in the class NF.

The important property of G is the absence of torsion. Thus G can be a candidate to solve some other open questions like the Kadison–Kaplanski conjecture about idempotents or a torsion free version of the Atiyah conjecture on L^2-Betti numbers.

One of the main criterion concerning amenability is a probabilistic criterion of Kesten, which is related to spectral properties of the Markov operator on groups. In [20], some spectral properties of this operator on the group G are investigated, although there is no full solution concerning the problem of amenability.

The notion of branch groups was introduced in [17]. Besides the classical examples of branch groups given in [12, 14, 24], a number of recent examples of branch type had appeared in [2, 4, 5].

In [5], a torsion free just non-solvable group H generated by a 4 state automaton is investigated. It shares many common properties with our group G and indeed can be embedded in pro-finite completion of G. Nevertheless in some respects G allows a deeper study than H. For instance, the growth type of H is unknown as well as the question about solvability of conjugacy problem while for G it is known that it has exponential growth and the conjugacy problem is solvable [20].

More comprehensive investigation of both groups H and G is an actual task for the future.

2. Preliminaries

2.1. *Automata groups*

The automata that we use are finite invertible automata with the same input and output alphabet $D = \{0, 1, \ldots, d-1\}$ for some $d > 1$. Such an automaton A has a finite set Q of states, transition and exit functions: $\phi : D \times Q \to Q$, $\psi : D \times Q \to D$ and is characterized by the quadruple $\langle D, Q, \phi, \psi \rangle$.

The automaton A is invertible if for any $q \in Q$ the function $\psi(\cdot, q) : D \to D$ is a bijection and therefore can be identified with the corresponding element σ_q of the symmetric group S_d on $d = \#D$ symbols, where $\#$ denotes cardinality. We consider the right action of S_d on D.

There is a convenient way to describe a finite automaton by a labeled graph $\Gamma(A)$ whose vertices correspond to the elements of Q, two states $q, s \in Q$ are joined by an oriented edge labeled by $i \in D$ if $\phi(i, q) = s$, and each vertex $q \in Q$ is labeled by the corresponding element σ_q of the symmetric group.

The automata just defined are non-initial automata. To make them initial one has to declare some state $q \in Q$ as the initial state. The initial automaton $A_q = \langle D, Q, \phi, \psi, q \rangle$ operates on the right on finite and infinite sequences over D in the following way. For each input symbol $x \in D$ it immediately gives the output $y = \psi(x, q)$ and changes its initial state to $\phi(x, q)$. Then it takes the next input symbol and repeats the procedure for the initial automaton $A_{\phi(x,q)}$ and so on. Joining the output of A_q with the input of another automaton $B_s = \langle D, S, \alpha, \beta, s \rangle$ one gets a map which corresponds to a finite automaton $C_{(q,s)}$ called the composition of A_q and B_s and denoted by $A_q \star B_s$.

This automaton can formally be described as the automaton with the set of states $Q \times S$ and the transition and exit functions Φ, Ψ defined as

$$\Phi(i, (x, y)) = (\phi(i, x), \alpha(\psi(i, x), y)),$$

$$\Psi(i, (x, y)) = \beta(\psi(i, x), y),$$

with the initial state (q, s).

The composition $A \star B$ of two non-initial automata is defined by similar formulas for transition and exit functions only without indicating the initial state.

Two initial automata are called equivalent if they determine the same map on the set of strings. There is an algorithm for minimization of a finite automata which produces an automaton with minimal number of states which is equivalent to the given one [10]. The elements of an automaton group are equivalence classes of automata, i.e. automorphisms of trees (see Sec. 2.2).

An automaton producing the identity map on the set of strings is called trivial. If A is an invertible automaton then for each state q the initial automaton A_q has an inverse automaton A_q^{-1} such that $A_q \star A_q^{-1}$, $A_q^{-1} \star A_q$ are equivalent to a trivial automaton, i.e. an automaton inducing the identity map on the set of sequences.

The inverse automaton can formally be defined as the automaton $\langle D, Q, \tilde{\phi}, \tilde{\psi}, q\rangle$ where $\tilde{\phi}(i,s) = \phi(i^{\sigma_s^{-1}}, s)$, $\tilde{\psi}(i,s) = i^{\sigma_s^{-1}}$ for $s \in Q$. The classes of equivalence of finite invertible automata over the alphabet D constitute a group which is called a finite automata group (it depends on D). Any set of initial automata generates some subgroup of this group.

Now let A be an invertible non-initial automaton as in the beginning. Let $Q = \{q_1, \ldots, q_t\}$ be the set of states of Q and let $A_{q_1}, \ldots, A_{q_t}$ be the set of initial automata which can be created from A. The group $G(A) = \langle A_{q_1}, \ldots, A_{q_t}\rangle$ is called the group determined or generated by A (see [15]).

2.2. *Action on a tree*

The strings over an alphabet $D = \{0, \ldots, d-1\}$ are in one-to-one correspondence with vertices of a d-regular rooted tree T_d whose root vertex corresponds to the empty string.

An initial automaton A_q acting on strings over D acts also on T_d by automorphisms. Thus for any group generated by an automata, in particular for a group of the form $G(A)$ there is a canonical action on the corresponding tree. This action is described in more details in [19, 23].

2.3. *Automata groups and wreath products*

There is a close relation between automata groups and wreath products [10]. For a group of the form $G(A)$ it has the following interpretation.

Let $q \in Q$ be a state of A and let $\sigma_q \in S_d$ be the label of this state. For each symbol $i \in D$ denote by $A_{q,i}$ the initial automaton with the initial state $\phi(q,i)$ (thus $A_{q,i}$ for $i = 0, 1, \ldots, d-1$ runs through the set of initial automata A_p for which there is an edge from q to p in the graph of the automaton A.

Let G and F be finitely generated groups such that F is a permutation group of a set X (the case which will be interesting for us is when F is a symmetric group S_d and X is the set $\{0, 1, \ldots, d-1\}$). We assume that F acts on X on the right. Define the wreath product $G \wr F$ of these groups as follows. Elements of $G \wr F$ are couples (g, γ) where $g : X \to G$ is a function such that $g(x)$ is different from the identity element id_G of G only for finitely many elements x in X, and where γ is an element of F. The multiplication in $G \wr F$ is defined as follows:

$$(g_1, \gamma_1)(g_2, \gamma_2) = (g_3, \gamma_1\gamma_2)$$

where

$$g_3(x) = g_1(x)g_2(x^{\gamma_1}) \quad \text{for } \gamma \in F\,.$$

We will write the elements of the group $G \wr S_d$ as expressions $(a_0, \ldots, a_{d-1})\sigma$, where $a_0, \ldots, a_{d-1} \in G$ and $\sigma \in S_d$.

The group $G = G(A)$ embeds in the wreath product $G \wr S_d$ via the map

$$A_q \to (A_{q,0}, \ldots, A_{q,d-1})\sigma_q\,,$$

where $q \in Q$. This can be seen from the action of $G(A)$ on the tree, described in the previous section (for more on this see [23, Sec. 3.3]). The expression on the right hand side of the arrow will be called a wreath decomposition of A.

We will write in the text $A_q = (A_{q,0}, \ldots, A_{q,d-1})\sigma_q$ without making special comments.

2.4. *Branch and fractal groups*

For a group $G = G(A)$ acting by automorphisms on a corresponding rooted tree T we denote by $St_G(n)$ the subgroup of G consisting of those elements of G which act trivially on the level n in the tree T. Analogously for a vertex $u \in T$ we denote by $St_G(u)$ the subgroup of G consisting of those elements of G which act trivially on u, i.e. fixing u. If an element $g \in G$ belongs to the stabilizer $St_G(1)$ of the first level, then because G embeds in the wreath product $G \wr S_d$ we get an embedding $\phi : St_G(1) \to G \times \cdots \times G$ (d times) in the base group of the wreath product. This defines canonical projections $\psi_i : St_G(1) \to G$ ($i = 1, \ldots, d$) on the ith coordinate.

A stabilizer $St_G(n)$ of the nth level is the intersection of all stabilizers of this level. For any vertex $u \in T$ we can define the projection $\psi_u : St(u) \to G$.

Definition 1. A group G is called fractal if for any vertex u, $\psi_u(St_G(u)) = G$ after the identification of the tree T with a subtree T_u with a root u.

A rigid stabilizer of the vertex u is the subgroup $Rist_G(u)$ of automorphisms of G that act trivially on the complement of the subtree T_u. A rigid stabilizer of the nth level $Rist_G(n)$ is the group generated by rigid stabilizers of vertices on this level.

A spherically transitive group $G \leq \mathrm{Aut}(T)$ is called a branch group if $Rist_G(n)$ is a subgroup of finite index for every $n \in \mathbb{N}$. A spherically transitive group $G \leq \mathrm{Aut}(T)$ is called a weakly branch group if $|Rist_G(n)| = \infty$ for every $n \in \mathbb{N}$.

As long as there is no confusion we will omit the subscript G in $St_G(u)$, $Rist_G(u)$, etc.

Definition 2. We say that the group G is regular weakly branch over a subgroup $K \neq \{1\}$ if $K \geq K \times \cdots \times K$ (d factors each of which acts on a corresponding subtree Tu, $|u| = 1$).

The embedding $G \to G \wr S_d$, $g \to (g_0, \ldots, g_{d-1})\sigma$ defines a restriction g_i of g in the vertex i of the first level. The iteration of this procedure leads to the notion of restriction g_u of g at any vertex u.

The length of a word w and of the element g is denoted by $|w|$ and $|g|$ respectively.

Definition 3. A group G is called contracting if there is $\lambda < 1$ and $C, L \in \mathbb{N}$ such that for any vertex u at the level $l > L$

$$|g_u| < \lambda|g| + C\,.$$

We use the notations $x^y = y^{-1}xy$, $[x, y] = x^{-1}y^{-1}xy$ and $\langle X\rangle^Y$ for the normal closure of X in Y.

The proof of Theorem 1 will be splitted into several steps and is given in Secs. 3–5.

3. Algebraic Properties of G

Let G be the group given by the automaton from Fig. 1, i.e. $G = \langle a, b\rangle$ where $a = (1, b)$ and $b = (1, a)\varepsilon$, ε is the non-unit element of S_2 and we use notations of Sec. 2.3.

Proposition 1. *The group G is fractal.*

Proof. We have

$$St_G(1) = \langle a, a^b, b^2\rangle\,.$$

But

$$\begin{aligned} a &= (1, b) \\ a^b &= \varepsilon(1, a^{-1})(1, b)(1, a)\varepsilon = (b^a, 1) \qquad (1) \\ b^2 &= (a, a)\,, \end{aligned}$$

and each of two projections of $St_G(1)$ is G, i.e. G is fractal. □

Proposition 2. *The group G is regular weakly branch over G', i.e.*

$$G' \geq G' \times G'\,.$$

Proof. Indeed, as

$$[a, b^2] = (1, [b, a])$$

using the fractalness of G we get $G' \geq \langle[a, b^2]\rangle^G \geq 1 \times \langle[b, a]\rangle^G = 1 \times G'$ and $(1 \times G')^b = G' \times 1$. Therefore G' contains $G' \times G'$ and as $G' \neq 1$, the group G is regular weakly branch over G'. □

Lemma 1. *Let w be a word in $\{a, b, a^{-1}, b^{-1}\}$ representing the identity in G. Then the exponent sum of both a and b must be* 0.

Proof. Let us prove it by induction on the length of $|w| = n$.

For $n = 1, 2$ by taking if necessary the inverse or conjugation of w we have to consider only the cases a, b, a^2, b^2 and $a^{\pm 1}b^{-1}$ which are non-trivial elements of G.

Suppose that Lemma 1 is true for all words of length $n \geq 2$. Let us consider the word w of length $n+1$. If w is a non-zero power of a or b then w represents a non-trivial element of G. If w contains both a and b by taking if necessary the inverse or conjugation of w we can suppose that w ends with $a^{\pm 1}b^{-1}$. As w represents the trivial element and $a^{\pm 1}b^{-1} = (a^{-1}, b^{\pm 1})\varepsilon$ we have

$$w = (w_0, w_1)\varepsilon(a^{-1}, b^{\pm 1})\varepsilon = (w_0 b^{\pm 1}, w_1 a^{-1}) .$$

Because w represents the trivial element, $w_0 b^{\pm 1}$ and $w_1 a^{-1}$ represent the identity elements as well. But their lengths are strictly less than the length of w so by induction the exponent sum of a and b in $w_0 b^{\pm 1}$ and $w_1 a^{-1}$ is zero. As $a = (1, b)$ and $b = (1, a)\varepsilon$ the exponent sum of a in w is equal to the exponent sum of b in the word $w_0 b^{\pm 1} w_1 a^{-1}$ and similarly for a. Thus the exponent sum of a and b in w is zero which finishes the proof. □

For an element $x \in G$ let $|x|_a$ denote the exponent sum of a in x. By Lemma 1, this does not depend on a word representing x.

Lemma 2. *A pair (x, y) with $x, y \in G$ determines an element of $St_G(1)$ if and only if $|x|_a = |y|_a$.*

Proof. As $St_G(1)$ is generated by $a = (1, b)$, $a^b = (b^a, 1)$ and $b^2 = (a, a)$, it is clear that if $(x, y) \in St_G(1)$ then $|x|_a = |y|_a$. Let us show the converse. Because $(1, b)$ and (a, a) belong to $St_G(1)$, for any element $y \in G$ we have $(a^{|y|_a}, y) \in G$. Similarly as $(b, 1) \in St_G(1)$ for any $x \in G$ we have $(x, a^{|x|_a}) \in G$. Thus for any $x, y \in G$ such that $|x|_a = |y|_a$, because $(a^n, a^n) \in St_G(1)$ we get

$$St_G(1) \ni (a^{|y|_a}, y)(a^{-|y|_a}, a^{-|y|_a})(x, a^{|x|_a}) = (x, y) .$$ □

Let $\phi : St_G(1) \to G \times G$ be the embedding from Sec. 2.4 and let $A = \langle a \rangle^G$ and $B = \langle b \rangle^G$.

Lemma 3.

$$Im\phi = (B \times B) \rtimes \langle (a, a) \rangle .$$

Proof. This follows from the fact that $b^2 = (a, a)$, $a = (1, b)$, $a^b = (b^a, 1)$ and fractalness of G. □

Proposition 3. *The group G is torsion-free.*

Proof. If there is an element of finite order in G (the order must be a power of 2) then there is an element z of order 2. Using fractalness we can assume that $z \notin St_G(1)$. Then $z = (x, y)b$ for some x, y in G. Since $(x, y) \in St_G(1)$ it can be written as a word of even b exponent and so one can write xy as a word of even a exponent. Thus by Lemma 1, xya is not the identity and this contradicts the fact that

$$\begin{aligned} z^2 &= ((x,y)(1,a)\varepsilon)^2 = ((x,ya)\varepsilon)^2 \\ &= (xya, yax) = 1 \end{aligned}$$

and proves (c). □

Proposition 4. *The groups G is of exponential growth.*

Proof. This is a consequence of the following.

Lemma 4. *The semi-group generated by a and b is free.*

Proof. Let us consider two different words $U(a,b)$ and $V(a,b)$ which represent the same element and such that $\rho = \max(|U|,|V|)$ is minimal. Direct verification shows that ρ cannot be 0 or 1.

Let us also assume that $|U|_b$ the number of occurrences of b in U is even (and so $|V|_b$). If this is not the case we can consider the words bU and bV thus enlarging ρ by 1.

Now U and V are products of

$$a^m = (1, b^m)$$

and of

$$ba^m b = (1,a)\varepsilon(1,b^m)(1,a)\varepsilon = (b^m a, a)\,.$$

Thus if in one of these words there is no b, say $U = a^m$, after projecting U and V on the first coordinate we obtain $1 = V_0$ where V_0 is a projection on the first coordinate and is a non-empty word and $|V_0| < |V| \leq \rho$. This gives a contradiction with minimality of U and V.

We consider now the situation when b appears in both words at least twice. Suppose that in both words b appears twice. If originally the number of occurrences of b in U and V was one, then because of their minimality they have to be equal to ba^n and $a^m b$. But $ba^n = (b^n, a)\varepsilon$ and $a^m b = (1, b^m a)\varepsilon$ which means that if these words are different they represent different elements.

Thus either both words contain two b's and $|U|,|V| \leq \rho$ or one of them contains at least four b's and $|U|,|V| \leq \rho + 1$.

If we consider the projections of U and V on the second coordinate we obtain two different words (because of minimality of U and V they have to end with different letters) and of strictly smaller length. This gives a desired contradiction with minimality of ρ.

This proves (d) because if the semi-group is free the growth is exponential. □

Lemma 5. *If* $g \in G'$, $g = (g_0, g_1)$ *then* $g_0 g_1 \in G'$.

Proof. Indeed $G' = \langle [a,b] \rangle^G$, $[a,b] = (b^a, b^{-1})$ and $b^a b^{-1} \in G'$. $\square$

Lemma 6. *We have*

$$G/G' = \mathbb{Z}^2 .$$

Proof. We need to prove that for any $i, j \in \mathbb{Z}$, $(i,j) \neq (0,0)$, we have $a^i b^j \notin G'$. Every element in G' can be written as a word of exponent 0 in both a and b. Thus if $a^i b^j \in G'$ with $(i,j) \neq (0,0)$ we get a contradiction with Lemma 1. $\square$

Lemma 7. *We have*:

$$G' = (G' \times G') \rtimes \langle c \rangle \tag{2}$$

where $c = [a,b]$.

Proof. As $G' = \langle [a,b] \rangle^G$, $[a,b] = ([a,b^{-1}]b, b^{-1})$ and $G' \geq G' \times G'$ we get that the element $d = (b, b^{-1})$ belongs to G'. Therefore

$$G' = (G' \times G')\langle d \rangle = (G' \times G')\langle c \rangle ,$$

because as it can be easily seen $c^a, c^{a^{-1}}, c^b, c^{b^{-1}} \in \langle c \rangle \bmod G' \times G'$. Finally $\langle c \rangle \cap (G' \times G') = 1$ as $c = (b^a, b^{-1})$ and $b^n \in G'$ if and only if $n = 0$. $\square$

Lemma 8. *The following relation holds*:

$$\gamma_3(G) = (\gamma_3(G) \times \gamma_3(G)) \rtimes \langle [[a,b],b] \rangle .$$

Proof. We start with relations

$$\gamma_3(G) = \langle [[a,b],a], [[a,b],b] \rangle^G ,$$

$$[[a,b],a] = [(b^a, b^{-1}), (1,b)] = 1 ,$$

$$[[a,b],b] = (b^{-a}, b)\varepsilon(1, a^{-1})(b^a, b^{-1})(1,a)\varepsilon = (b^{-a}, b)(b^{-a}, b^a) = (b^{-2a}, bb^a) , \tag{3}$$

the first two of which allow to conclude that

$$\gamma_3(G) = \langle [[a,b],b] \rangle^G .$$

Because of the relation

$$[a, b^2] = (1, [a,b])$$

we get

$$[[a,b^2],a] = [(1,[a,b]),(1,b)] = (1, [[a,b],b]) .$$

Let $\xi = [[a,b],b]$. Direct computations show that $\xi^a, \xi^{a^{-1}}, \xi^b, \xi^{b^{-1}} \in \langle\xi\rangle$ mod $\gamma_3(G) \times \gamma_3(G)$ and $\langle\xi\rangle \cap (\gamma_3(G) \times \gamma_3(G)) = 1$ because of (3) and the fact that $(bb^a)^n \in G'$ if and only if $n = 0$ and $\gamma_3(G) \le G'$. □

Lemma 9. *The following relation holds*:

$$G'' = \gamma_3(G) \times \gamma_3(G)\,.$$

Proof. Let $f = (1,c) \in G$. We have

$$G'' \ni [f, d^{-1}] = [(1,[a,b]),(b^{-1},b)] = (1,[[a,b],b])\,.$$

This implies that $G'' \ge 1 \times \gamma_3(G)$ and therefore $G'' \ge \gamma_3(G) \times \gamma_3(G)$. As $G'' \le \gamma_3(G)$ in any group G by Lemma 8 we only need to prove that $\langle\xi\rangle \cap G'' = 1$. Using the relation (3) we get from Lemma 7 that

$$G'' = \langle [c,f]\rangle^G\,.$$

But

$$[c,f] = [(b^a, b^{-1}),(1,c)] = [1,[b^{-1},[a,b]]] \in 1 \times \gamma_3(G)\,.$$

This ends the proof. □

As far as (g) is concerned, the word problem is solvable for any finitely generated group of finite automata (for instance one can multiply generators and minimize the product using the algorithm of minimization of finite automata). For groups of branch type there is a very effective branch algorithm solving the word problem which was first discovered for the group from [12]. This algorithm works also for the group G which we consider.

Proposition 5. *The word problem is solvable.*

Proof. Let w be a word on the alphabet a, b, a^{-1}, b^{-1}.

(1) Check if $w \in St_G(1)$ (otherwise $w \neq^G 1$).
(2) Compute $w = (w_0, w_1)$. Then

$$w =^G 1$$

if and only if $w_i =^G 1$, $i = 0, 1$. Go to 1. Replacing w by w_0, w_1 and proceed with each w_0, w_1 as with w. If in some step one gets a word different from the identity in G then $w \neq 1$ in G. If in some step n all words $w_{i_1}, \ldots, w_{i_n}$ are empty then $w = 1$ in G.

This algorithm converges because if $g = (g_0, g_1) \in \langle a\rangle^G$ then $g_0, g_1 \in \langle b\rangle^G$ and we have the reduction of the length. □

Proposition 6. *The group G is just non-solvable.*

Proof. This will be a consequence of the following

Proposition 7. *Let G be a regular weakly branch over K. Then for any normal subgroup $N \triangleleft G$ there is n such that*

$$K'_n < N$$

where $K_n = K \times \cdots \times K$ (d^n times and each factor acts on the corresponding subtree).

Proof. The proof is similar to the proof of Theorem 4 from [17, p. 149]. □

Lemma 10. *Let G be a regular weakly branch over K (i.e. $K \triangleleft G$, $K > K \times \cdots \times K$ (d times), $K \neq 1$) and G/K, $K/(K \times \cdots \times K)$ are solvable. Then G' is just non-solvable.*

Proof. G/K_n is solvable because we have the exact sequence

$$1 \to K_{n-1}/K_n \to G/K_n \to G/K_{n-1} \to 1$$

in which G/K_{n-1}, $K_{n-1}/K_n = (K/(K \times \cdots \times K))^{d^{n-1}}$ are solvable and induction arguments work. We also have the exact sequence

$$1 \to K_n/K'_n \to G/K'_n \to G/K_n \to 1$$

where K_n/K'_n, G/K_n are solvable.

If $n \neq 1$ is normal in G then N contains K'_n for some n. Therefore G/K'_n can be mapped onto G/N and as G/K'_n is solvable G/N is solvable as well. □

As a corollary we obtain that G is just non-solvable. Indeed G is weakly branch over G' and $G'/G' \times G' \sim \mathbb{Z}$.

4. A Presentation for G

We say that a group G has a finite L presentation if there exists a finite alphabet S, finite sets of reduced words Q, R and an endomorphism $\phi : F_S \to F_S$ such that G is isomorphic to a group with a following presentation

$$\langle S; Q, \phi^n(R), n = 0, 1, \ldots \rangle .$$

The first such a presentation was obtained by I. Lysionok [27] for the group from [12] (see also [16]). Finite L-presentations are useful for embedding into finitely presented groups [16].

We will obtain a presentation for the group G which is of L-type but unfortunately is not finite. Here we follow the strategy of getting such a type of presentation from [16].

Let $F_2 = \langle a, b\rangle$ be the free group and $\Delta = \langle b^2, a\rangle^{F_2}$. Then $\Delta = \langle a, a^b, b^2\rangle$ is a free group of rank 3 which is a subgroup of index 2 in F_2. Define $\Psi : \Delta \to F_2 \times F_2$ as

$$\Psi(a) = (1, b) = x\,,$$

$$\Psi(a^b) = (b^a, 1) = y\,,$$

$$\Psi(b^2) = (a, a) = z\,,$$

and let $\tilde{\Delta} = \Psi(\Delta)$.

Proposition 8. *The group $\tilde{\Delta}$ admits the following presentation*

$$\tilde{\Delta} = \langle x, y, z | [x^{z^m}, y^{z^n}] = 1, m, n \in \mathbb{Z}\rangle\,.$$

Proof. Let $R = \langle b\rangle^{F_2}$. Then $R \times R \leq \tilde{\Delta}$ as $(1, b), (b^a, 1) \in \tilde{\Delta}$ and $p_i(\tilde{\Delta}) = F_2$ (p_i for $i = 0, 1$ are projections of $\tilde{\Delta}$ into factors). As $R \times R = \langle x, y\rangle^{F_2\times F_2}$, it is clear that we have the exact sequence

$$1 \to R \times R \to \tilde{\Delta} \to C \to 1$$

where $C = \langle(\bar{a}, \bar{a})\rangle = \langle\bar{z}\rangle$ ($\bar{a}$, $\bar{z}$ means the image of the elements a and z).

Let us show that $\langle z\rangle \cap (R \times R) = (1, 1)$. For this it is enough to show that $\langle a\rangle \cap \langle b\rangle^{F_2} = 1$. But $\langle b\rangle^{F_2}$ is a free group generated by $a^{-m}ba^m$, $m \in \mathbb{Z}$ and it is clear that $a^n \notin \langle b\rangle^{F_2}$ for every $n \neq 0$. So $\tilde{\Delta} = (R \times R) \rtimes \langle z\rangle$ is a semi-direct product.

Let us use for $m \in \mathbb{Z}$ the following notations

$$b_m = a^{-m}ba^m \in F_2\,,$$

$$b_m^+ = (b_m, 1) \in F_2 \times F_2\,,$$

$$b_m^- = (1, b_m) \in F_2 \times F_2\,.$$

Then $R \times R$ can be described by a presentation

$$\langle b_m^-, b_m^+ | [b_m^-, b_n^+] = 1, m, n \in \mathbb{Z}\rangle\,.$$

A presentation for $\tilde{\Delta}$ (as $\tilde{\Delta}$ is a semi-direct product) is

$$\tilde{\Delta} = \langle b_m^-, b_n^+, z | (b_m^-)^z = b_{m+1}^-, (b_m^+)^z = b_{m-1}^+, (b_m^-)^{z^{-1}}$$

$$= b_{m-1}^-, (b_m^+)^{z^{-1}} = b_{m+1}^+, [b_m^-, b_n^+] = 1, m, n \in \mathbb{Z}\rangle$$

as can be easily checked by a verification of the action of z on generators of $R \times R$.

And in generators x, y, z we get a simpler presentation

$$\tilde{\Delta} = \langle x, y, z, | (x^{z^m})^z = x^{z^{m+1}}, (y^{z^m})^z = y^{z^{m+1}}, [x^{z^m}, y^{z^m}] = 1, m, n \in \mathbb{Z}\rangle$$

$$= \langle x, y, z, | [x^{z^m}, y^{z^m}] = 1, m, n \in \mathbb{Z}\rangle\,. \qquad \square$$

From this follows that

$$\mathrm{Ker}\Psi = \langle [a^{b^{2m}}, a^{b^{2n+1}}], m, n \in \mathbb{Z}\rangle^{\Delta}\,.$$

Indeed KerΨ is normal in F_2 because

$$\tau_{m,n}^{b} = [a^{b^{2m+1}}, a^{b^{2(n+1)}}] = \tau_{n+1,m}^{-1}\,,$$

$$\tau_{m,n}^{b^{-1}} = [a^{b^{2m-1}}, a^{b^{2n}}] = \tau_{n,m-1}^{-1}$$

where $\tau_{m,n} = [a^{b^{2m}}, a^{b^{2n+1}}]$. Therefore

$$\mathrm{Ker}\Psi = \langle [a^{b^{2m}}, a^{b^{2n+1}}], m,n \in \mathbb{Z}\rangle^{F_2} = \langle [a, a^{b^k}], k = 1,3,5,\ldots\rangle^{F_2}\,.$$

Proposition 9. *The group G has the following presentation*:

$$G = \langle a, b | \sigma^k(\tau_m) = 1, m = 2l+1, l = 0,1,\ldots, k = 0,1,\ldots\rangle\,,$$

where

$$\tau_m = [a^{b^m}, a]$$

and

$$\sigma : \begin{cases} a \to b^2 \\ b \to a \end{cases}$$

Proof. Let $G = F_2/\Omega$ and let Δ be the above subgroup of index two in F_2. Let p_i be the projection on the ith coordinate, $i = 0,1$.

Lemma 11. *The following relation holds*

$$p_1\Psi\sigma = id\,.$$

Proof. We have

$$p_1\Psi\sigma(a) = p_1\Psi(b^2) = p_1(a,a) = a\,,$$

$$p_1\Psi\sigma(b) = p_1\Psi(a) = p_1(1,b) = b\,.$$

□

Lemma 12. *If $g \in F_2'$ then $\Psi\sigma(g) = (1,g)$.*

Proof. Because

$$\Psi\sigma(a) = (a,a)\,,$$

$$\Psi\sigma(b) = (1,b)\,,$$

for any commutator word $w(a,b)$ we have

$$\Psi\sigma(w(a,b)) = (w(a,1), w(a,b)) = (1, w(a,b))$$

as $w(a,1)$ freely reduces to the empty word. □

Let

$$\Omega_0 = \mathrm{Ker}\Psi = \langle \tau_m, m = 2l+1, l = 0,1,\ldots\rangle^{F_2}\,,$$

$$\Omega_n = \{w \in F_2 : w \in \Delta, w_i = p_i\Psi(w) \in \Omega_{n-1}, i = 0,1\}\,.$$

Lemma 13. *Ω_n is a normal subgroup in F_2.*

Proof. Of course Ω_n is a subgroup. We will prove by induction on n that if $w \in \Omega_n$ then $w^{a^{\pm 1}}, w^{b^{\pm 1}} \in \Omega_n$. We write $w = (w_0, w_1)$ instead of $w_i = p_i \Psi(w)$.

It is clear that Ω_0 is normal in F_2.

If $w \in \Delta$, $w = (w_0, w_1)$ then

$$w^{a^{\pm 1}} = (w_0, w_1^{\pm b}),$$

$$w^{b^{\pm 1}} = (w_1^{\pm a}, w_0)$$

and by inductive assumption $w_0, w_1^{\pm b}, w_1^{\pm a} \in \Omega_{n-1}$. □

Lemma 14. *The following relation holds*

$$\Omega = \bigcup_{n=1}^{\infty} \Omega_n .$$

Proof. This follows from the algorithm solving the word problem in G. □

Lemma 15. *We have*

$$\Omega_n = \langle \sigma^k(\tau_m), m = 2l+1, l = 0, 1, \ldots, k = 0, 1, \ldots, n \rangle^{F_2} .$$

Proof. We will prove this by induction on n.

For $n = 0$, this is correct by definition of Ω_0.

Let us check that $\Psi(\Omega_{n+1}) = \Omega_n \times \Omega_n$, $n \geq 0$. Indeed by definition of Ω_{n+1} we have $\Psi(\Omega_{n+1}) \leq \Omega_n \times \Omega_n$ and

$$\Psi \sigma^{i+1}(\tau_m) = (1, \sigma^i(\tau_m)), i \geq 0 \tag{4}$$

$$\Psi(b^{-1}\sigma^{i+1}(\tau_m) b) = (a^{-1}\sigma^i(\tau_m) a, 1) . \tag{5}$$

If

$$\Omega_n = \langle \sigma^k(\tau_m), m \in \mathbb{Z}, k = 0, 1, \ldots, n \rangle^{F_2} ,$$

then

$$\Omega_{n+1} = \langle \sigma^k(\tau_m), m \in \mathbb{Z}, k = 0, 1, \ldots, n+1 \rangle^{F_2} ,$$

as from (4) and (5), it follows that $\langle \sigma^k(\tau_m), m \in \mathbb{Z}, k = 0, 1, \ldots, n+1 \rangle^{F_2}$ maps onto $\Omega_n \times \Omega_n$.

This ends the proof of Proposition 9. □

Proposition 10. *σ induces an injective endomorphism $G \to G$ and therefore G embeds into the group*

$$\tilde{G} = \langle a, b, s | a^s = b^2, b^s = a, \tau_m = 1, m = 2l+1, l = 0, 1, \ldots \rangle$$

which is an ascending HNN-extention of G.

Proposition 11. *The group G has the presentation*

$$G = \langle a, b | \sigma^k(\theta^m([a, a^b])) = 1, m = 0, 1, \ldots, k = 0, 1, \ldots \rangle$$

where

$$\theta : \begin{cases} a \to a^{b^2+1} \\ b \to b \end{cases} .$$

Proof. Let $n = 1, 2, \ldots$ and

$$U_n = \langle \tau_1, \tau_3, \ldots, \tau_{2n-1} \rangle^{F_2}$$

$$V_n = \langle \theta^i(\tau_1), i = 0, \ldots, n-1 \rangle^{F_2} .$$

Proposition 11 is a consequence of the following lemma

Lemma 16. *The relation $U_n = V_n$ holds for $n = 1, 2, \ldots$.*

Proof. Let us show the inclusion $U_n \subset V_n$ by induction on n.

For $n = 1$, the statement is obvious.

Let us suppose that $U_n \subset V_n$. Now

$$\begin{aligned} \theta^n(\tau_1) &= [a^{(b^2+1)^n}, a^{(b^2+1)^n b}] \\ &= [a \cdot a^{nb^2 + \cdots + nb^{2(n-1)} + b^{2n}}, a^{b^2 + nb^4 + \cdots + nb^{2n}} \cdot a^{b^{2(n+1)}}] \\ &= a^{-(nb^2 + \cdots + nb^{2(n-1)} + b^{2n})} a^{-1} a^{-b^{2(n+1)}} a^{-(b^2 + nb^4 + \cdots + nb^{2n})} \\ &\quad \times a a^{nb^2 + \cdots + nb^{2(n-1)} + b^{2n}} a^{b^2 + nb^4 + \cdots + nb^{2n}} a^{b^{2(n+1)}} . \end{aligned} \tag{6}$$

So using the relations from U_n we can make the cancellations in the above product and get

$$[a, a^{b^{2n+1}}] = \tau_{n+1} \in V_{n+1} .$$

The opposite inclusion $U_n \supset V_n$ follows immediately, because using the relators from U_n we can cancel to the empty word the right hand side of (6). $\square$

This finishes the proof of Proposition 11.

Proposition 12. *The group G has the following presentation*

$$G = \langle a, b | \sigma^\varepsilon(\theta^m([a, a^b])) = 1, m = 0, 1, \ldots, \varepsilon = 0, 1 \rangle .$$

Proof. We have the relations

$$\sigma^{2k}(a) = a^{2k} ,$$

$$\sigma^{2k}(b) = b^{2k} ,$$

$$\sigma^{2k-1}(a) = b^{2k} ,$$

$$\sigma^{2k-1}(b) = a^k .$$

and by Lemma 16

$$\langle \tau_1, \tau_3, \ldots, \tau_{2m-1} \rangle^{F_2} = \langle \theta^i(\tau_1), 0 \le i \le m-1 \rangle^{F_2}\,.$$

Thus the relator

$$\sigma^{2k-1}(\tau_m) = [b^{2k}, (b^{2k})^{a^{km}}]$$

is a corollary of the relator

$$[b^2, b^{2\cdot a^{km}}] = \sigma(\tau_{km})\,.$$

Similarly

$$\sigma^{2k}(\tau_m) = [a^{2k}, (a^{2k})^{b^{2km}}]$$

is a corollary of

$$[a, a^{b^{2km}}] = \tau_{2km}$$

and this finishes the proof of Proposition 12. □

Remark 1. θ is not an endomorphism of G. But it is still possible that G has a finitely presented HNN extension.

5. Amenability Properties

Let SG_0 be the class of groups whose all finitely generated subgroups have subexponential growth. Assume that $\alpha > 0$ is an ordinal and that we have defined EG_β for each ordinal $\beta < \alpha$. Then if α is a limit ordinal, set

$$SG_\alpha = \bigcup_{\beta<\alpha} SG_\beta$$

and if α is not a limit ordinal, set SG_α is the class of groups which can be obtained from groups in $SG_{\alpha-1}$ by applying either extensions or direct limits. Let

$$SG = \bigcup_{\alpha} SG_\alpha\,.$$

Groups from this class are called subexponentially amenable groups. The class SG is closed with respect to taking subgroups and quotients (see [31, Theorem 21]).

Proposition 13. *The group G is not subexponentially amenable, i.e. $G \notin SG$.*

Proof. We start with the following lemmas

Lemma 17. *The following relation holds*

$$p_0(\gamma_3(G)) = \langle \gamma_3(G), b^{2a} \rangle\,.$$

Proof. It follows from Lemma 8 and from the relation (3). □

Lemma 18. *We have*

$$p_0(\langle\gamma_3(G), b^{2a}\rangle) = \langle\gamma_3(G), b^{2a}, a\rangle\,.$$

Proof. This follows from the previous lemma and the relation $b^{2a} = (a, a^b)$. □

Lemma 19. *For the projection on the second coordinate we have*

$$p_1(\langle\gamma_3(G), b^{2a}, a\rangle) = G\,.$$

Proof. This follows from Lemma 8 and the relations $b^{2a} = (a, a^b)$ and $a = (1, b)$. □

Assume $G = SG_\alpha$ and α minimal. α cannot be zero as G has exponential growth. α is not a limit ordinal because if $G \in SG_\alpha$ for some limit ordinal then $G \in SG_\beta$ for some ordinal $\beta < \alpha$. Also α is not a direct limit (of an infinite increasing sequence of groups) because it is finitely generated. Therefore there exist N, $H \in SG_{\alpha-1}$ such that the following exact sequence holds

$$1 \to N \to G \to H \to 1\,.$$

By Proposition 7, there exists n such that $N > (Rist_G(n))' \geq G'' \times \cdots \times G''$ (2^n times). Therefore $G'' \in SG_{\alpha-1}$ and hence $\gamma_3(G) \in SG_{\alpha-1}$ as follows from Lemma 9. Each class SG_α is closed with respect to taking homomorphic images and subgroups [31]. From Lemmas 17–19, it follows that $G \in SG_{\alpha-1}$. Contradiction.

This finishes the proof of (h).

We are going to prove (i) and (j). We start with

Lemma 20. (a) *Let $g \in St_G(1), g = (g_0, g_1)$. Then*

$$|g_0| + |g_1| \leq |g| \tag{7}$$

and $|g_i| < |g|$ $i = 0, 1$ if $g \neq 1$ and

$$g \notin \{a^n, b^{-1}a^n b, ba^n b^{-1}, n \in \mathbb{Z} \setminus 0\} = X\,.$$

(b) *Let $g \notin St(1)$, $g = (g_0, g_1)\varepsilon$. Then $|g_0| + |g_1| \leq |g|$ and $|g_i| < |g|$, $i = 0, 1$ if*

$$g \notin \{a^n b, b^{-1}a^n, n \in \mathbb{Z}\} = Y\,.$$

Proof. Both (a) and (b) will by proven by induction on length $|g|$. First of all we prove (a).

For g such that $|g| = 1$, this is obvious.

Let $|g| > 1$ and g is represented by a word w such that $|g| = |w|$. In the word w let us take the longest possible among the following set of prefixes: $Z = \{b^{-1}a^n b, ba^n b^{-1}, b^{-1}a^n b^{-1}, ba^n b, a^n, b^{2n}\}$. Such a nonempty prefix must exist.

Denote the part of the word w after this prefix by w'. We can apply the induction on the length of $|w|$ because of relations

$$\begin{aligned}
b^{-1}a^n b &= (a^{-1}b^n a, 1)\,,\\
b a^n b^{-1} &= (a b^n a^{-1}, 1)\,,\\
b^{-1}a^n b^{-1} &= (a^{-1}b^n, a^{-1})\,,\\
b a^n b &= (a b^n, b)\,,\\
a^n &= (1, b^n)\,,\\
b^{2n} &= (a^n, a^n)
\end{aligned} \tag{8}$$

which lead to (7).

Let us prove the second part of (a). If $w' \notin X$ then by inductive assumption $|w_i| < |w|$, where $w = (w_0, w_1)$, and so $|g_i| < |g|$ for $i = 0, 1$. If $w' \in X$ and is nonempty then because of the relations (we take all possible products of elements of X and Z that do not give a free cancellation)

$$\begin{aligned}
a^n \cdot b^{-1}a^k b &= (a^{-1}b^k a, b^n)\,,\\
a^n \cdot b a^k b^{-1} &= (a b^k a^{-1}, b^n)\,,\\
b^{-1}a^k b \cdot a^n &= (a^{-1}b^k a, b^n)\,,\\
b a^k b^{-1} \cdot a^n &= (a b^k a^{-1}, b^n)\,,\\
b a^k b^{-1} \cdot b^{-1}a^n b &= (a b^k a^{-1} a^{-1} b^n a, 1)\,,\\
b^{-1}a^k b \cdot b a^n b^{-1} &= (a^{-1}b^k a a b^n a^{-1}, 1)\,,\\
b^{2k} \cdot a^n &= (a^k, a^k b^n)\,,\\
b^{2k} \cdot b a^n b^{-1} &= (a^k a b^n a^{-1}, a^k)\,,\\
b^{-1}a^k b^{-1} \cdot a^n &= (a^{-1}b^k a^{-1}, b^n)\,,\\
b^{-1}a^k b^{-1} \cdot b^{-1}a^n b &= (a^{-1}b^k a^{-1} b^n a, a^{-1})\,,\\
b a^k b \cdot a^n &= (a b^k, b b^n)\,,\\
b a^k b \cdot b a^n b^{-1} &= (a b^k a b^n a^{-1}, b)\,,
\end{aligned}$$

we have the reduction of length at each coordinate and thus we can apply the induction assumption.

Now let us prove (b).

For g of length 1, the statement is obvious.

Let $|g| > 1$, $g = (g_0, g_1)\varepsilon$ and $g =^G w$ (this notation means that the word w represents g in G), $|g| = |w|$. Then w is equal to $a^n w', b^{2n} w'$ or $b^{\pm 1} a^k w'$ for some w' and $n, k \geq 1$. In order to prove (7) for $w = b^{\pm 1} a^k w'$ we use relations (8),

$ba^k = (b^k, a)\varepsilon$ and $b^{-1}a^k = (a^{-1}b^k, 1)\varepsilon$. For other cases we use the relations

$$a^n = (1, b^n)\,,$$
$$b^{2n} = (a^n, a^n)$$

which show that the induction can be applied. Thus in all cases we get (7).

Let us prove the second part of (b). If $w = b^{\pm 1}a^k$ then $w' \in St_G(1)$. In this case if $w' \notin X$ then we have the reduction of length by (a) and by $ba^k = (b^k, a)\varepsilon$ and $b^{-1}a^k = (a^{-1}b^k, 1)\varepsilon$. If $w' \in X$ then because of

$$b^{\pm 1}a^k \cdot b^{-1}a^n b = b^{\pm 1}(a^{-1}b^n a, b^k)\,,$$
$$b^{\pm 1}a^k \cdot ba^n b^{-1} = b^{\pm 1}(ab^n a^{-1}, b^k)$$

we have the reduction of length at each coordinate and thus we can apply the induction assumption.

If $w = a^n w'$ or $b^{2n}w'$ then $w' \notin St_G(1)$. If $w' \in Y$ and has the form $b^{-1}a^k$ or $a^k b$ then

$$a^n \cdot b^{-1}a^k = (a^{-1}b^k, b^k)\varepsilon\,,$$
$$b^{2n} \cdot a^k b = (a^{n+1}, a^n b^k)\varepsilon\,,$$

and again we can apply the induction assumption. □

Proposition 14. *The group G does not contain a free group of rank 2, i.e. $G \in NF$.*

Proof. Let $g, h \in G$, $g, h \neq 1$. We have to construct a freely reduced word w over the alphabet $\{g, h, g^{-1}, h^{-1}\}$ such that $w(g, h) = 1$ in G.

This will be done by induction on the number $|g| + |h|$.

If $|g| + |h| = 2$ then $g = a^{\pm 1}, h = b^{\pm 1}$ and using the relation $[a, a^b] = 1$ for the pair a, b or similar relations for other pairs we get the base of induction.

Let $|g| + |h| > 2$. We have to consider four cases, depending on whether g, h belong to $St(1)$ or not. By symmetry this can be reduced to three cases.

Case I. $g, h \in St(1)$, $g = (x, y), h = (u, v)$.

Then $|x|, |y| \leq |g|$ and $|u|, |v| \leq |h|$.

Thus

$$\begin{aligned} |x| + |u| &\leq |g| + |h| \\ |y| + |v| &\leq |g| + |h|\,. \end{aligned} \tag{9}$$

This leads to two subcases.

Case I(a). If g or $h \notin X$ we have strict inequalities in (9) by Lemma 20.

In this case by induction there exist non empty words w_1, w_2 such that $w_1(x, u) =^G 1$ and $w_2(y, v) =^G 1$. But

$$\begin{aligned} w_1(g, h) &= (w_1(x, u), w_1(y, v)) = (1, w_1(y, v)) \\ w_2(g, h) &= (w_2(x, u), 1)\,. \end{aligned} \tag{10}$$

The word $w(g,h) = [w_1(g,h), w_2(g,h)]$ represents the unit element in G' and is a nontrivial word after a free reduction if w_1 and w_2 are not powers of the same word.

Case I(b). If $w_1 = u^k$, $w_2 = u^l$ for some word u and $k,l \in \mathbb{Z}$, $k,l \neq 0$ then $u^{kl}(g,h) =^G 1$. If $g,h \in X$ then $w(g,h)$ can be constructed explicitly. Namely

$$[a^n, b^{-1}a^n b] = 1$$

$$[a^n, ba^n b^{-1}] = 1\,.$$

For the words $b^{-1}a^n b, ba^n b^{-1}$ we have $b^{-1}a^n b = (a^{-1}b^n a, 1)$, $ba^n b^{-1} = (ab^n a^{-1}, 1)$ so it is enough to find a non-trivial word w such that $w(a^{-1}b^n a, ab^n a^{-1}) =^G 1$ and we reduce the problem to the previous Case I(a) because $a^{-1}b^n a, ab^n a^{-1} \notin X$.

Case II. $g \in St(1), h \notin St(1), g = (x,y), h = (u,v)\varepsilon$. Assume $g \notin X$. Take $h^2 = (uv, vu)$. Then $|uv| \leq |u| + |v| \leq |h|$, $|vu| \leq |h|$. Moreover

$$\begin{aligned} |x| + |uv| &< |g| + |h| \\ |y| + |vu| &< |g| + |h|\,. \end{aligned} \tag{11}$$

We can apply the same arguments, as in the Case I, replacing gh by g, h^2.

If $g \in X$ then $g = (1,y)$ or $(y,1)$ where $y \in \{b^n, a^{-1}b^n a, ab^n a^{-1}\}$ and

$$|y| + |uv| \leq |g| + |h|\,.$$

So the problem is to find w_1 such that $w_1(y, uv) = 1$. If $y \in St(1), uv \in St(1)$ we can proceed as in the Case I.

If $y \notin St(1), uv \in St(1)$ and $uv \notin X, |uv| = |h|$ we can do as before.

If $uv \in X$ then we have the following relation

$$[y \cdot uv \cdot y^{-1}, uv] = 1\,,$$

which can be easily checked.

Case III. $g,h \notin St(1), g = (x,y)\varepsilon, h = (u,v)\varepsilon$.

Then

$$g^2 = (xy, yx), h^2 = (uv, vu)$$

and

$$\begin{aligned} |xy| + |uv| &\leq |g| + |h| \\ |yx| + |vu| &\leq |g| + |h|\,. \end{aligned} \tag{12}$$

If both inequalities in (12) are strict then w can be constructed as $[w_1, w_2]$ where $w_1(xy, uv) =_G 1$, $w_2(yx, vu) =_G 1$. If one of the inequalities in (12) is not strict then we cosider several cases.

Assume $xy \in St(1)$ or $uv \in St(1)$. Then this case reduces to the Cases I or II.

Assume $xy, uv \notin St(1)$. Then $\bar{x} \neq \bar{y} \bmod 2$ and $\bar{u} \neq \bar{v} \bmod 2$ for the images $\bar{x}$, $\bar{y}$, $\bar{u}$, $\bar{v}$ of x, y, u and v in $S_2 = G/St_G(1)$. Consider $gh = (xv, yu)$ and $hg = (uy, vx)$ for which we have the inequalities

$$|xv| + |uy| \leq |x| + |y| + |u| + |v| \leq |g| + |h| \,.$$

If xv or $yu \in St(1)$ we can proceed as in the Cases I or II. If not, we have the relations $\bar{x} \neq \bar{v} \bmod 2$ and $\bar{y} \neq \bar{u} \bmod 2$. This implies that $\bar{x} = \bar{u} \bmod 2$ and $\bar{y} = \bar{v} \bmod 2$.

But now

$$\begin{aligned} gh^{-1} &= (xu^{-1}, yv^{-1}) \,, \\ h^{-1}g &= (v^{-1}y, u^{-1}x) \,, \\ |xu^{-1}| + |v^{-1}y| &\leq |g| + |h| \end{aligned}$$

and $xu^{-1}, yv^{-1}, v^{-1}y, u^{-1}x \in St(1)$ so we can reduce considerations to the Case I. □

This finishes the proof of (i).

Proposition 15. *The group G is contracting with parameters $\lambda = \frac{2}{3}, C = 1, L = 1$.*

Proof. All words of length two up to taking the inverse are b^2, ab, ba, a^2, $a^{-1}b$, ab^{-1}. For these words except from $a^{-1}b$

$$\begin{aligned} b^2 &= (a, a) \\ ab &= ((1,1), (b,a)\varepsilon)\varepsilon \\ ba &= (b, a)\varepsilon \\ a^2 &= ((1,1), (a,a)) \\ ab^{-1} &= (a^{-1}, b)\varepsilon \end{aligned} \tag{13}$$

and we have reduction of the length on the second level by $\frac{1}{2}$. As far the word $a^{-1}b$ is concerned we have

$$\begin{aligned} (a^{-1}b)b &= (a, b^{-1}a) \\ (a^{-1}b)a^{\pm 1} &= (b^{\pm 1}, b^{-1}a)\varepsilon \,. \end{aligned} \tag{14}$$

Therefore the statement holds for all words of length ≤ 2 and moreover for words on the left hand side of (14), we have the reduction of the length by $\frac{2}{3}$. Up to adding or removing one letter, every word can be written using a product of words from (13) and (14). Thus for $C = 1$, we get the contraction with coefficient $\lambda = \frac{2}{3}$. □

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